

Frozenset data type

A **Frozen Set** is an **immutable version of a set**.

1. Collection of unique elements enclosed within frozenset(iterable) is called a frozenset.
2. Duplicate values are NOT allowed.
3. Unordered collection (no fixed order).
4. Indexing not supported.
5. Slicing not supported.
6. Different data types allowed.
7. Immutable in nature (elements can be added or removed).

```
fs1 = frozenset([10, 20, 30])
fs2 = frozenset((1, 2, 3))
fs3 = frozenset({"Python", "Java", "C++"})
```

```
print(fs1)
print(fs2)
print(fs3)
```

Output:-

```
frozenset({10, 20, 30})
frozenset({1, 2, 3})
frozenset({'Python', 'Java', 'C++'})
```

Duplicate Values in Frozen Set:-

```
fs = frozenset([10, 20, 20, 30, 10])
print(fs)
```

Output:-

```
frozenset({10, 20, 30})
```

Creating Empty Frozen Set:-

```
fs = frozenset()
```

```
print(fs)
print(type(fs))
```

output:-

```
frozenset()
<class 'frozenset'>
```

Python In-built functions for Frozen Set:-

Step	Syntax	Explanation
1	len()	Number of elements
2	type()	Check datatype
3	id()	Memory address
4	frozenset()	Create frozen set
5	print()	Print output

```
fs = frozenset([10, 20, 30])
```

```
print(len(fs))
print(type(fs))
print(id(fs))
```

Output:-

```
3
<class 'frozenset'>
135446907837043
<class 'frozenset'>
135446907837040
```

Frozen Set Methods:

1. union()
2. intersection()
3. difference()
4. symmetric_difference()
5. issubset()
6. issuperset()
7. isdisjoint()

1. Union



```
A = frozenset{1,2,3,4,5,6,7}
B = frozenset{5,6,7,8,9,10}
print(A.union(B))
O/P:--
frozenset{1,2,3,4,5,6,7,8,9,10}
```

2 Intersection



```
A = frozenset{1,2,3,4,5,6,7}
B = frozenset{5,6,7,8,9,10}
print(A.intersection(B))
O/P:--
frozenset{5,6,7}
```

3. Difference



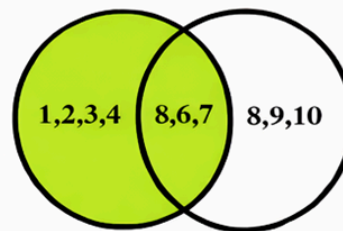
```
A = frozenset{1,2,3,4,5,6,7}
B = frozenset{5,6,7,8,9,10}
print(A.difference(B))
O/P:--
frozenset{1,2,3,4}
```

4. Symmetric_difference



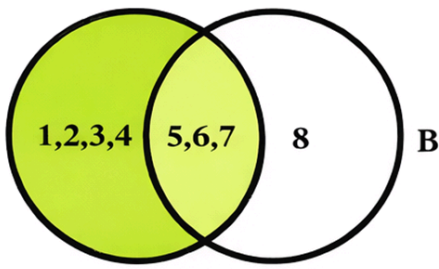
```
A = frozenset{1,2,3,4,5,6,7}
B = frozenset{5,6,7,8,9,10}
print(A.symmetric_difference(B))
O/P:--
frozenset{1,2,3,4,8,9,10}
```

4. Symmetric_difference(B)



```
A = frozenset{1,2,3,4,5,6,7}
B = frozenset{5,6,7,8,9,10}
print(A.symmetric_difference(B))
O/P:--
frozenset{1,2,3,4,8,9,10}
```

1. isdisjoint



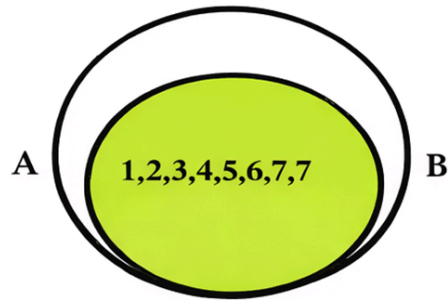
A = frozenset{{1,2,3,4}}

B = frozenset{{5,6,7}}

print(A.isdisjoint(B))

O/P:—

2. issubset



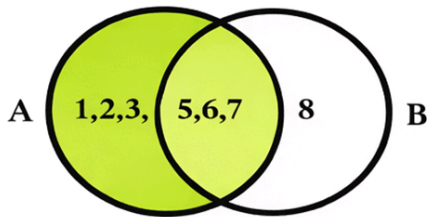
A = frozenset{{1,2,3,4}}

B = frozenset{1,2,3,4,5,6,7,7}

print(A.issubset(B))

O/P:— True

3. issuperset



A = frozenset{{1,2,3,4,5,6,7}}

B = frozenset{{5,6,7}}

print(A.issuperset(B))

O/P:— True